

# P6 — Validation + QA + Scientific Evaluation

SIH 26143 · Detailed 6-Phase Work Plan · 6 owned work packages

## ROLE OBJECTIVE

Own quantitative validation and scientific QA so every major claim has a defined test and every failure is visible.

## HOW TO USE THIS PLAN

Each package has four parts: what to build/do, the concrete deliverable, dependencies, and an acceptance condition. The owner is responsible for making the output usable by the next phase—not merely completing a task locally.

### PHASE 1 — DATA & PREPROCESSING

#### Work package 1: Dataset audit

Audit image/mask correspondence, class/label structure, file integrity, metadata, train/validation/test separation and possible leakage; document what the dataset actually contains and what it does not contain.

**Deliverable:** dataset audit sheet + approved split manifest.

**Dependencies / acceptance:** Depends on dataset files. Acceptance: split is fixed before final metric reporting.

### PHASE 2 — SPILL DETECTION

#### Work package 2: Detection evaluation

Build the evaluation script for IoU, Dice, precision, recall and false-positive analysis; evaluate held-out data; create representative error cases and distinguish segmentation errors from look-alike confusion.

**Deliverable:** reproducible detection metrics + error gallery.

**Dependencies / acceptance:** Depends on P1 model. Acceptance: metrics can be regenerated from the saved checkpoint and split.

### PHASE 3 — SOURCE RECONSTRUCTION

#### Work package 3: Source-recovery validation

Generate controlled synthetic release/source scenarios; run backward reconstruction; measure corridor/time-window containment of the known source; test sensitivity to forcing uncertainty and observation perturbations.

**Deliverable:** source-recovery test set + containment/error report.

**Dependencies / acceptance:** Depends on P5 reconstruction pipeline. Acceptance: ground-truth source and release time are known by construction.

### PHASE 4 — AIS CORRELATION

#### Work package 4: AIS-gap testing

Create controlled AIS deletion scenarios; remove observations at different gap patterns; rerun correlation and measure changes in candidate set/ranking; verify missing data does not receive adverse evidence weight.

**Deliverable:** AIS robustness report.

**Dependencies / acceptance:** Depends on P4 correlation pipeline. Acceptance: behaviour under missing AIS is quantified and documented.

### PHASE 5 — EVIDENCE FUSION + UI

#### Work package 5: Fusion evaluation

Compare evidence-family behaviour individually and in combination; test ranking stability, sensitivity to missing evidence and abstention thresholds; ensure results are explainable and reproducible.

**Deliverable:** evidence-fusion evaluation matrix.

**Dependencies / acceptance:** Depends on P2/P4/P5 integration. Acceptance: every fusion test has a defined input scenario and expected behaviour.

### PHASE 6 — VALIDATION & DEPLOYMENT

**Work package 6: Final validation + QA report**

Run the complete matrix across detection, look-alikes, source recovery, AIS gaps, attribution scenarios, abstention, reproducibility and failure handling; freeze reported metrics; document limitations and unresolved risks.

**Deliverable:** final QA report + validation checklist + demo sign-off.

**Dependencies / acceptance:** Depends on all team outputs. Acceptance: every major claim in the PPT maps to a test, measured result or explicit planned item.

## CROSS-TEAM HANDOFF RULE

No component is considered complete if its output cannot be consumed by the next owner. Preserve IDs, timestamps, coordinate systems, provenance, uncertainty and failure states across every handoff.

Core project rule: AIS gaps, uncertain metocean forcing, ambiguous SAR signatures or weak evidence must increase uncertainty or trigger abstention—not be converted into fabricated confidence.